UC SAN DIEGO / ECE 251C

7
Multiresolution analysis and wavelet filtering

Orthogonal decompositions, two-scale equations, and fast analysis

How can discrete filters compute projections of a continuous-domain signal?

7.1 From local averages to a discrete algorithm

In the previous lecture, we built a fine approximation from a coarse approximation and a detail. For Haar wavelets, this meant replacing two neighboring values by their average and their difference. We now explain why these simple operations are instances of a more general construction.

There are two descriptions to connect. The spaces V_i and W_i describe *functions*: the approximation at resolution i and the detail added at the next resolution. A computer instead stores *coefficient sequences*. Our goal is to obtain the coarse and detail coefficients directly from the fine coefficients, without returning to the continuous-domain signal.

Recall the geometric decomposition. If $P_i f$ is the orthogonal projection of f onto V_i, then

$$V_{i+1} = V_i \oplus W_i, \qquad P_{i+1} f = P_i f + (P_{i+1} f - P_i f).$$

The difference belongs to W_i and is orthogonal to V_i. Thus the finer approximation contains both the old information and one new increment.

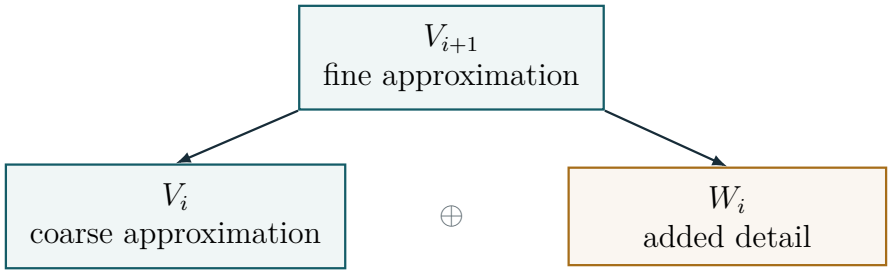


Figure 7.1: One finer representation, split into two orthogonal components. The two-scale expansions will turn these projections into filtering operations.

7.2 What defines a multiresolution analysis?

The Haar construction provides a model, but the approximation functions need not be boxes. We retain the relations among the spaces and state them independently of that example.

Definition (Mallat and Meyer, ca. 1988). An *orthonormal multiresolution analysis* (MRA) is a nested sequence of closed subspaces

$$\cdots \subset V_{-1} \subset V_0 \subset V_1 \subset \cdots \subset L^2(\mathbb{R})$$

with the following properties:

1. **Fine resolutions are complete:**

$$\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{L^2} = L^2(\mathbb{R}).$$

   Any finite-energy signal can be approximated arbitrarily well at sufficiently fine resolution.

2. **No information persists at every coarse scale:**

$$\bigcap_{i\in\mathbb{Z}} V_i = \{0\}.$$

   No nonzero finite-energy signal belongs to all the spaces.

3. **Changing scale changes resolution:**

$$f \in V_0 \quad \Longleftrightarrow \quad f(2^i \cdot) \in V_i.$$

   Compressing the coordinate by a factor of two moves up one level.

4. **Integer translates stay in the reference space:**

$$f \in V_0 \quad \Longrightarrow \quad f(\cdot - n) \in V_0, \qquad n \in \mathbb{Z}.$$

5. **One scaling function supplies the coordinates.** There is a function $\phi \in V_0$ such that $\{\phi(\cdot - n)\}_{n\in\mathbb{Z}}$ is an orthonormal basis of V_0.

Here *closed* means that an L^2 limit of functions in the space remains in the space. This ensures that orthogonal projection is defined. The closure in the first condition allows approximation; a signal need not lie in any one V_i.

For Haar, $\phi = \mathbf{1}_{[0,1)}$ and V_i consists of finite-energy functions constant on cells of width 2^{-i}. Other constructions use nonorthogonal stable bases (Riesz bases) or redundant stable families (frames). We keep the orthonormal setting here.

7.3 Detail spaces are orthogonal increments

An approximation space contains a complete description up to its resolution. A detail space contains only what is added in moving to the next resolution. It is therefore a *complement*, not a set difference:

$$W_i \coloneqq \{q \in V_{i+1} : \langle q, v\rangle_{L^2} = 0 \text{ for every } v \in V_i\}. \tag{7.1}$$

Equivalently, $W_i = V_{i+1} \cap V_i^{\perp}$. Orthogonality here uses

$$\langle f, g\rangle_{L^2} = \int_{\mathbb{R}} f(t)g^*(t)\,\mathrm{d}t, \qquad \|f\|_{L^2}^2 = \langle f, f\rangle_{L^2}.$$

As before, the *second* function is conjugated.

Why does this give a direct sum? For $v \in V_{i+1}$, write

$$v = P_i v + (v - P_i v).$$

The first term belongs to V_i. The second belongs to V_{i+1} and is orthogonal to V_i, so it belongs to W_i. A function in both V_i and W_i is orthogonal to itself and hence has zero norm. Thus the decomposition is unique:

$$\boxed{V_{i+1} = V_i \oplus W_i.} \tag{7.2}$$

The same projection argument as in Lecture 5 gives

$$\|v\|_{L^2}^2 = \|P_i v\|_{L^2}^2 + \|v - P_i v\|_{L^2}^2.$$

The two components divide the energy without duplicating it.

Different detail levels are orthogonal. If $k < i$, then

$$W_k \subset V_{k+1} \subset V_i, \qquad W_i \perp V_i.$$

Therefore $W_k \perp W_i$. A fine detail is orthogonal not only to its own coarse approximation, but also to every earlier detail already contained in that approximation.

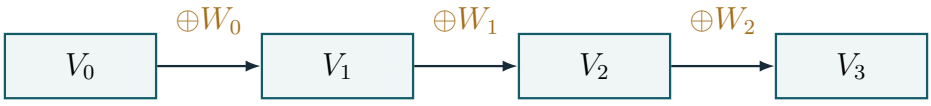


Figure 7.2: Approximation spaces are nested. Detail spaces are the mutually orthogonal increments between them; the detail spaces are not themselves nested.

7.4 Recovering the approximation spaces from the details

Repeatedly substitute $V_{k+1} = V_k \oplus W_k$. For integers $J < i$,

$$V_i = V_J \oplus W_J \oplus W_{J+1} \oplus \cdots \oplus W_{i-1}. \tag{7.3}$$

For example, $V_3 = V_0 \oplus W_0 \oplus W_1 \oplus W_2$. The upper detail index is 2, not 3: W_3 would supply the additional information needed to reach V_4.

The starting level J is arbitrary. Letting it move toward coarser resolutions suggests that all the earlier detail spaces should recover V_i:

$$\boxed{V_i = \overline{\bigoplus_{k=-\infty}^{i-1} W_k}^{L^2}.} \tag{7.4}$$

Here the direct sum consists of finite sums of elements from the mutually orthogonal spaces W_k; the closure includes their L^2 limits. The following argument explains why no component is missing.

Proof. Let U_i denote the right-hand side. Each W_k, $k < i$, lies in V_i, so $U_i \subset V_i$. For $v \in V_i$, let $r = v - \mathrm{proj}_{U_i} v$. Then $r \in V_i$ and $r \perp W_k$ for every $k < i$.

For any $J < i$, decompose r using equation (7.3). Orthogonality to $W_J, \ldots, W_{i-1}$ forces all its detail components to vanish. Hence $r \in V_J$ for every $J < i$. By nesting it also belongs to the spaces with $J \geq i$. The MRA intersection condition gives $r = 0$, so $v \in U_i$. □

This proof uses ordinary orthogonal projection onto a closed subspace: the residual is perpendicular to that subspace. The intersection condition rules out a residual shared by every coarse space.

Since the union of the V_i is dense, equation (7.4) also gives

$$L^2(\mathbb{R}) = \overline{\bigoplus_{i\in\mathbb{Z}} W_i}^{L^2}. \tag{7.5}$$

Alternatively, stop at the fixed reference resolution 0 and keep its approximation:

$$\boxed{L^2(\mathbb{R}) = V_0 \oplus \overline{\bigoplus_{i=0}^{\infty} W_i}^{L^2}.} \tag{7.6}$$

The space V_0 collects all the details from negative resolutions. It replaces them as a group; it does not eliminate their information.

7.5 From subspaces to wavelet synthesis

The scaling function gives an orthonormal basis at every resolution:

$$\phi_{i,n}(t) := 2^{i/2}\phi(2^i t - n), \qquad n \in \mathbb{Z}.$$

To describe the detail spaces, choose a wavelet ψ such that $\{\psi(\cdot - n)\}_{n\in\mathbb{Z}}$ is an orthonormal basis of W_0, and set

$$\psi_{i,n}(t) := 2^{i/2}\psi(2^i t - n).$$

The Haar wavelet from Lecture 6 has this property. For a general construction, merely choosing $\psi \in W_0$ is not enough: its translates must span W_0 and be orthonormal.

Why does the same normalization work at every scale? For any two functions u, v, the substitution $s = 2^i t$ gives

$$\int_{\mathbb{R}} 2^{i/2}u(2^i t)\, 2^{i/2}v^*(2^i t)\, \mathrm{d}t = 2^i 2^{-i} \int_{\mathbb{R}} u(s)v^*(s)\, \mathrm{d}s.$$

Thus normalized dilation preserves inner products. It maps V_0 onto V_i and V_1 onto V_{i+1}, so it also maps their orthogonal complements W_0 and W_i onto one another. The dilated families remain orthonormal bases.

Coefficients and reconstructed components. Define

$$a_i[n] := \langle f, \phi_{i,n}\rangle_{L^2}, \qquad d_i[n] := \langle f, \psi_{i,n}\rangle_{L^2}. \tag{7.7}$$

We write $d_i[n]$ for the detail coefficients called $b_i[n]$ in Lecture 6. To distinguish the sequence from a function, write $q_i(t)$ for the reconstructed detail:

$$P_i f = \sum_{n\in\mathbb{Z}} a_i[n]\phi_{i,n}, \qquad q_i = \mathrm{proj}_{W_i} f = \sum_{n\in\mathbb{Z}} d_i[n]\psi_{i,n}.$$

Square brackets denote coefficients; t is the continuous coordinate.

The mutually orthogonal spaces in equation (7.6) show that

$$\{\phi_{0,n} : n \in \mathbb{Z}\} \;\cup\; \{\psi_{i,n} : i \geq 0,\ n \in \mathbb{Z}\}$$

is an orthonormal basis of $L^2(\mathbb{R})$. The full signal is

$$\boxed{f(t) = \sum_{n\in\mathbb{Z}} a_0[n]\phi_{0,n}(t) + \sum_{i=0}^{\infty}\sum_{n\in\mathbb{Z}} d_i[n]\psi_{i,n}(t).} \tag{7.8}$$

The equality is in L^2. This is *synthesis*: coefficients weight known functions, and the resulting components are added. Analysis is the reverse task of finding those coefficients. Notice that the detail sum begins at $i = 0$; otherwise the entire component in W_0 would be missing.

7.6 The two-scale coefficients are inner products

The inclusion $V_0 \subset V_1$ has a concrete consequence: the scaling function itself can be expanded in the finer basis. Likewise, $\psi \in W_0 \subset V_1$. Orthogonal projection therefore supplies both expansions:

$$\phi(t) = \sum_{n\in\mathbb{Z}} g_0[n]\phi_{1,n}(t), \qquad g_0[n] = \langle \phi, \phi_{1,n}\rangle_{L^2}, \tag{7.9}$$

$$\psi(t) = \sum_{n\in\mathbb{Z}} g_1[n]\phi_{1,n}(t), \qquad g_1[n] = \langle \psi, \phi_{1,n}\rangle_{L^2}. \tag{7.10}$$

The factor $\sqrt{2}$ is already part of $\phi_{1,n}(t) = \sqrt{2}\phi(2t-n)$. The sequences g_0, g_1 are coordinates in this normalized basis, not the heights of the functions in the plots.

Calculate the Haar coefficients by projection. Take

$$\phi = \mathbf{1}_{[0,1)}, \qquad \psi = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}.$$

Only the two finer boxes inside $[0,1)$ can overlap either function. Hence

$$g_0[0] = \sqrt{2}\int_0^{1/2} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}}, \qquad g_0[1] = \sqrt{2}\int_{1/2}^{1} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}},$$

$$g_1[0] = \sqrt{2}\int_0^{1/2} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}}, \qquad g_1[1] = \sqrt{2}\int_{1/2}^{1} (-1)\,\mathrm{d}t = -\frac{1}{\sqrt{2}}.$$

All other coefficients vanish. Thus

$$g_0[n] = \frac{\delta[n] + \delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n] - \delta[n-1]}{\sqrt{2}}. \tag{7.11}$$

This recovers the masks found by inspection in Lecture 6, but now explains where their values come from.

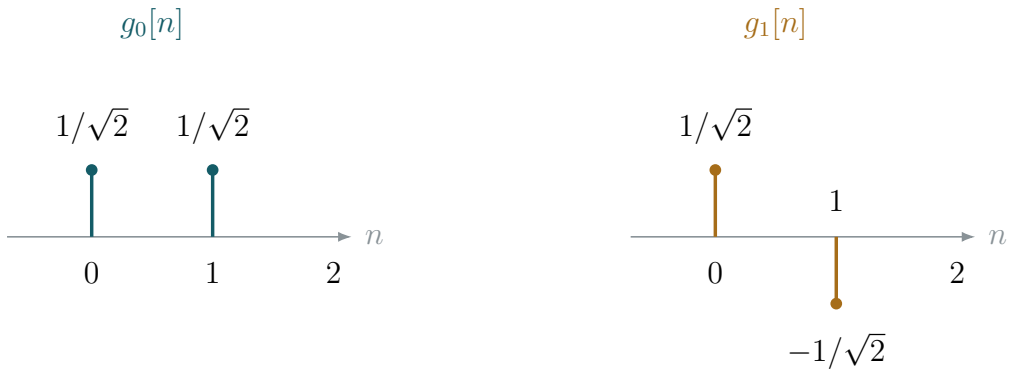


Figure 7.3: The same finer boxes build either an average or a difference. The minus sign in $g_1[1]$ comes from the negative half of the wavelet.

7.7 What information is available at the finest resolution?

Suppose our acquisition system has supplied the Haar coefficients at resolution 3:

$$a_3[n] = \langle f, \phi_{3,n} \rangle_{L^2}, \qquad f_3 = P_3 f = \sum_{n \in \mathbb{Z}} a_3[n] \phi_{3,n}.$$

These coefficients determine f_3, not necessarily f. The difference $f - f_3$ is information outside the acquired resolution. Our analysis will reorganize the information in f_3 without introducing additional measurements.

Coarser coefficients are already determined. Because $\phi_{2,m}$ and $\psi_{2,m}$ belong to V_3, both are orthogonal to $f - f_3$. Consequently

$$\langle f, \phi_{2,m} \rangle_{L^2} = \langle f_3, \phi_{2,m} \rangle_{L^2}, \qquad \langle f, \psi_{2,m} \rangle_{L^2} = \langle f_3, \psi_{2,m} \rangle_{L^2}.$$

We can therefore obtain the true coefficients $a_2[m], d_2[m]$ from f_3 alone.

Expand the fine representation inside the first inner product:

$$\begin{aligned} a_2[m] &= \left\langle \sum_{n \in \mathbb{Z}} a_3[n] \phi_{3,n}, \phi_{2,m} \right\rangle_{L^2} \\ &= \sum_{n \in \mathbb{Z}} a_3[n] \langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2}. \end{aligned} \tag{7.12}$$

The coefficient $a_3[n]$ is not conjugated: it appears in the *first* argument. In the same way,

$$d_2[m] = \sum_{n \in \mathbb{Z}} a_3[n] \langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2}. \tag{7.13}$$

Each output coefficient is a weighted sum of input coefficients. The weights measure overlaps between basis functions at two resolutions. For Haar, only two such overlaps survive.

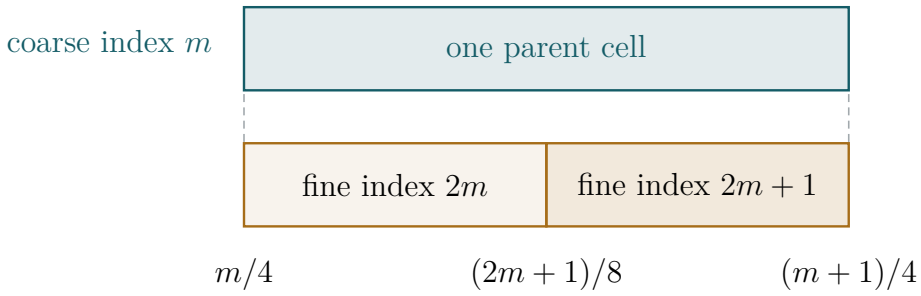


Figure 7.4: At resolutions 2 and 3, one parent interval contains two children. The factor $2m$ comes from expressing the parent location in the finer grid.

7.8 The low-pass weights: an overlap calculation

We now calculate the low-pass weight in equation (7.12), rather than guessing it from the averaging formula. The Haar functions are real, so conjugation has no effect in this calculation:

$$\langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2} = 2^{5/2} \int_{\mathbb{R}} \phi(8t - n)\phi(4t - m)\, \mathrm{d}t.$$

First put $u = 8t$, so $\mathrm{d}t = \mathrm{d}u/8$. Then put $v = u - 2m$, so $\mathrm{d}u = \mathrm{d}v$. The limits remain $-\infty$ and $+\infty$ in both changes of variable:

$$\begin{aligned} \langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2} &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi(u - n)\phi(u/2 - m)\, \mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi\big(v - (n - 2m)\big)\phi(v/2)\, \mathrm{d}v. \end{aligned} \tag{7.14}$$

The factor is $2^{5/2}/8 = 2^{-1/2}$. All dependence on the indices is now through the integer $r = n - 2m$.

The first box is one on $[r, r + 1)$; the second is one on $[0, 2)$. Their product integrates to the length of their overlap. That length is one for $r = 0, 1$ and zero for every other integer r.

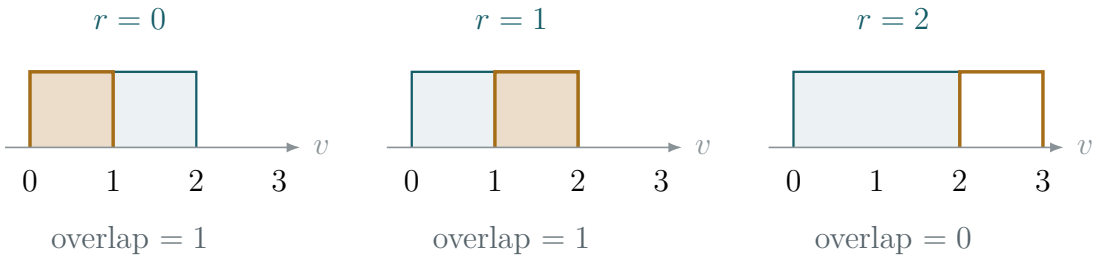


Figure 7.5: The broad box is $\phi(v/2)$; the narrow box is $\phi(v - r)$. A contact at one endpoint contributes zero to the integral.

Thus the overlap is exactly a shifted filter tap:

$$\langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2} = \begin{cases} 1/\sqrt{2}, & n = 2m \text{ or } 2m + 1, \\ 0, & \text{otherwise} \end{cases} = g_0[n - 2m]. \tag{7.15}$$

Substitution into equation (7.12) yields

$$a_2[m] = \sum_n a_3[n] g_0[n - 2m] = \frac{a_3[2m] + a_3[2m + 1]}{\sqrt{2}}.$$

The sum of neighboring coefficients is a consequence of overlapping basis functions.

7.9 The high-pass weights: signed overlaps

The detail coefficient uses a wavelet in place of the coarse box. Begin with the full integral:

$$\langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} = 2^{5/2} \int_{\mathbb{R}} \phi(8t - n)\psi(4t - m)\, \mathrm{d}t.$$

With $u = 8t$ and then $v = u - 2m$, the calculation is

$$\begin{aligned} \langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi(u - n)\psi(u/2 - m)\, \mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi\big(v - (n - 2m)\big)\psi(v/2)\, \mathrm{d}v. \end{aligned} \tag{7.16}$$

The prefactor is unchanged. The difference is that the broad function now has a sign:

$$\psi(v/2) = \begin{cases} 1, & 0 \le v < 1, \\ -1, & 1 \le v < 2, \\ 0, & \text{otherwise.} \end{cases}$$

For $r = n - 2m = 0$, the narrow box covers its positive half, giving integral 1. For $r = 1$, it covers the negative half, giving -1. All other integer translates have zero overlap.



Figure 7.6: The highlighted region is the product $\phi(v - r)\psi(v/2)$. Its signed area distinguishes the left child from the right child.

Consequently,

$$\langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} = \begin{cases} 1/\sqrt{2}, & n = 2m, \\ -1/\sqrt{2}, & n = 2m + 1, \\ 0, & \text{otherwise} \end{cases} = g_1[n - 2m]. \tag{7.17}$$

Putting these weights into equation (7.13) gives

$$d_2[m] = \sum_n a_3[n] g_1[n - 2m] = \frac{a_3[2m] - a_3[2m + 1]}{\sqrt{2}}.$$

Equal child coefficients give zero detail. For real signals, the sign records which child is larger: positive for the left, negative for the right.

7.10 Why convolution reverses the filters

Our projection formulas contain $g_k[n-2m]$. Standard convolution instead uses a tap indexed by *output location minus input location*:

$$(x * h)[\ell] = \sum_{n\in\mathbb{Z}} x[n]h[\ell - n].$$

The order of subtraction matters. To put the projection formulas into this form, define the real Haar analysis filters

$$\begin{aligned} h_0[k] &:= g_0[-k] = \frac{\delta[k] + \delta[k+1]}{\sqrt{2}}, \\ h_1[k] &:= g_1[-k] = \frac{\delta[k] - \delta[k+1]}{\sqrt{2}}. \end{aligned} \tag{7.18}$$

Then $h_k[2m-n] = g_k[n-2m]$, so

$$a_2[m] = \sum_n a_3[n]g_0[n-2m] = \sum_n a_3[n]h_0[2m-n] = (a_3 * h_0)[2m], \tag{7.19}$$

$$d_2[m] = \sum_n a_3[n]g_1[n-2m] = \sum_n a_3[n]h_1[2m-n] = (a_3 * h_1)[2m]. \tag{7.20}$$

Thus the analysis filters are the time-reversed two-scale masks. This reversal is required by the convolution convention; it is not a change in the projection being computed.

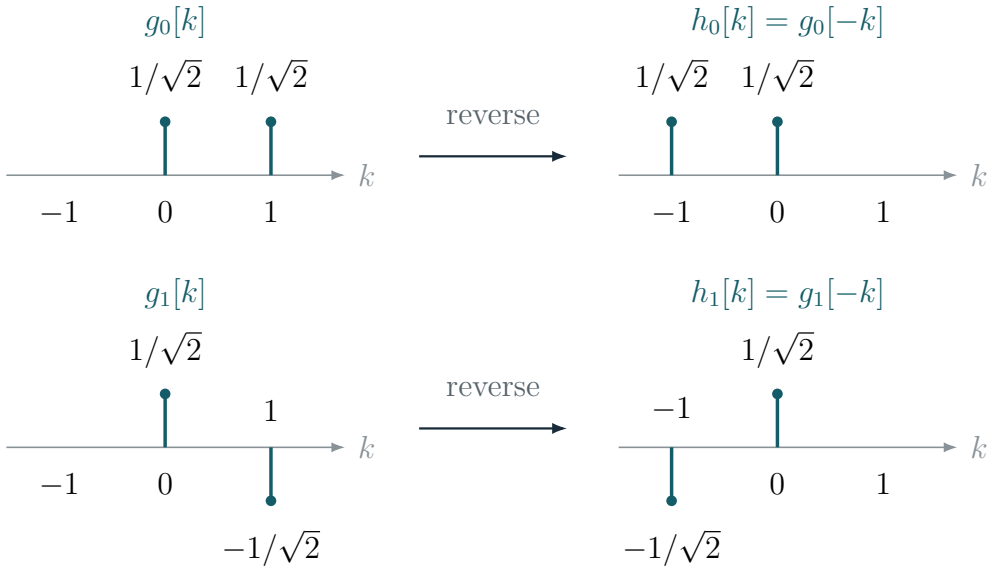


Figure 7.7: Reversal moves the tap at 1 to -1 without changing its value. In particular, the negative high-pass tap moves to the left of the origin.

The filters use neighboring entries $a_3[\ell]$ and $a_3[\ell+1]$. Both are already stored; the negative tap index requires no new signal sample.

7.11 Filtering followed by downsampling

Filtering produces a value at every fine-grid index. The projection needs only the values at the even indices $2m$, because the coarse grid has half as many locations per unit interval.

Definition. Downsampling a sequence z by two gives

$$((\downarrow 2)z)[m] := z[2m].$$

This retains the even-indexed entries and labels them by the coarse index m. It does not average neighboring entries. The averaging or differencing has already been done by the filter.

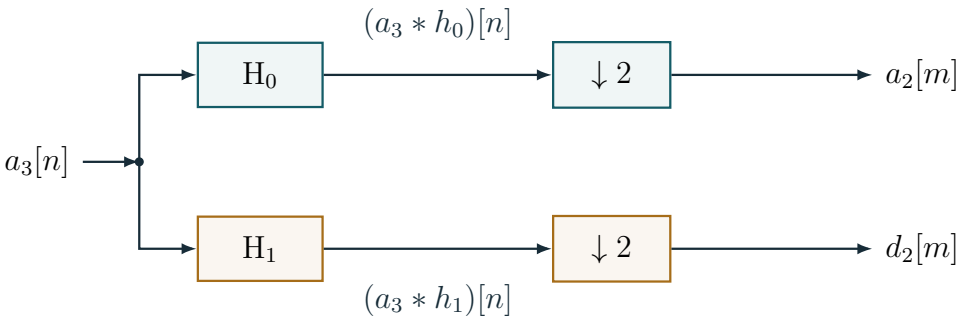


Figure 7.8: The two-channel analysis bank. The top branch computes approximation coefficients; the bottom computes detail coefficients. Each filter block is followed by its own downsampler.

For the Haar filters, write the two operations separately:

$$(a_3 * h_0)[n] = \frac{a_3[n] + a_3[n+1]}{\sqrt{2}}, \qquad (a_3 * h_0)[2m] = \frac{a_3[2m] + a_3[2m+1]}{\sqrt{2}},$$
$$(a_3 * h_1)[n] = \frac{a_3[n] - a_3[n+1]}{\sqrt{2}}, \qquad (a_3 * h_1)[2m] = \frac{a_3[2m] - a_3[2m+1]}{\sqrt{2}}.$$

The positions $2m, 2m+1$ are exactly the two children identified by the overlap calculations.

What has downsampling discarded? One branch alone is insufficient: a sum does not determine two numbers. The sum *and* difference do. Adding and subtracting the outputs gives

$$a_3[2m] = \frac{a_2[m] + d_2[m]}{\sqrt{2}}, \qquad a_3[2m+1] = \frac{a_2[m] - d_2[m]}{\sqrt{2}}.$$

Thus keeping both branches loses no information about f_3. For a finite array of even length, the two half-length outputs together contain the original number of coefficients.

7.12 The same filters work at every resolution

The overlap calculation used resolutions 3 and 2, but nothing depends on the number 3. The two-scale equations explain why the same masks can be reused.

Substitute $2^i t - m$ for t in equation (7.9) and multiply by $2^{i/2}$:

$$\begin{aligned}\phi_{i,m}(t) &= 2^{i/2} \sum_{k\in\mathbb{Z}} g_0[k]\sqrt{2}\,\phi\big(2(2^i t - m) - k\big) \\ &= \sum_{k\in\mathbb{Z}} g_0[k]\, 2^{(i+1)/2}\phi\big(2^{i+1}t - (2m+k)\big) \\ &= \sum_{k\in\mathbb{Z}} g_0[k]\phi_{i+1,2m+k}(t) = \sum_{n\in\mathbb{Z}} g_0[n-2m]\phi_{i+1,n}(t).\end{aligned}$$

The last equality is the index change $n = 2m + k$. Repeating the same dilation in equation (7.10) gives

$$\psi_{i,m} = \sum_{n\in\mathbb{Z}} g_1[n-2m]\phi_{i+1,n}.$$

Keep track of conjugation. Take the inner product of f with these expansions. A coefficient in the second argument is conjugated:

$$\begin{aligned}a_i[m] &= \sum_n g_0^*[n-2m]\, a_{i+1}[n], \\ d_i[m] &= \sum_n g_1^*[n-2m]\, a_{i+1}[n].\end{aligned} \tag{7.21}$$

For real Haar masks this is precisely the calculation above. If the two-scale masks are complex, the correct analysis filters include conjugation as well as reversal:

$$h_0[k] = g_0^*[-k], \qquad h_1[k] = g_1^*[-k]. \tag{7.22}$$

Therefore the general projection rule is

$$\boxed{a_i[m] = (a_{i+1} * h_0)[2m], \qquad d_i[m] = (a_{i+1} * h_1)[2m].} \tag{7.23}$$

The filters do not change with i. Changing level changes the meaning of one coefficient-grid step: it corresponds to a different distance in the continuous domain.

For an orthonormal scaling and wavelet construction, these formulas compute exact projections. Choosing two arbitrary filters does not, by itself, guarantee such a construction. Here the filters inherit their properties from the bases whose overlaps defined them.

7.13 A complete Haar analysis by hand

Continue the signal from Lecture 6 with values $(1, 3, 5, 7, 7, 7, 5, 5)$ on successive eighths of $[0, 1)$ and zero elsewhere. Because it lies in V_3, its acquisition is exact. A normalized basis function on an eighth-cell has height $\sqrt{8}$, so a cell value c gives coefficient $c/\sqrt{8}$:

$$a_3 = \frac{1}{\sqrt{8}}(1, 3, 5, 7, 7, 7, 5, 5).$$

Apply the pair formulas at each level. The first two pairs give

$$a_2[0] = \frac{1+3}{\sqrt{8}\sqrt{2}} = 1, \qquad d_2[0] = \frac{1-3}{\sqrt{8}\sqrt{2}} = -\frac{1}{2},$$
$$a_2[1] = \frac{5+7}{\sqrt{8}\sqrt{2}} = 3, \qquad d_2[1] = \frac{5-7}{\sqrt{8}\sqrt{2}} = -\frac{1}{2}.$$

The equal pairs $(7, 7)$ and $(5, 5)$ have zero details. Continuing yields

resolution i	a_i	d_i
2	$(1, 3, 7/2, 5/2)$	$(-1/2, -1/2, 0, 0)$
1	$(2\sqrt{2}, 3\sqrt{2})$	$(-\sqrt{2}, 1/\sqrt{2})$
0	(5)	(-1)

For example, $a_1[0] = (1+3)/\sqrt{2} = 2\sqrt{2}$ and $d_1[0] = (1-3)/\sqrt{2} = -\sqrt{2}$. At the final split, $a_0[0] = (2\sqrt{2} + 3\sqrt{2})/\sqrt{2} = 5$ and $d_0[0] = -1$.

These are *coefficients*, not cell heights. Substituting them into the synthesis formula gives

$$f = 5\phi - \psi - \sqrt{2}\psi_{1,0} + \frac{1}{\sqrt{2}}\psi_{1,1} - \frac{1}{2}\psi_{2,0} - \frac{1}{2}\psi_{2,1}. \tag{7.24}$$

No term was estimated: filtering has recovered the inner products of this function with the coarse and detail basis functions.

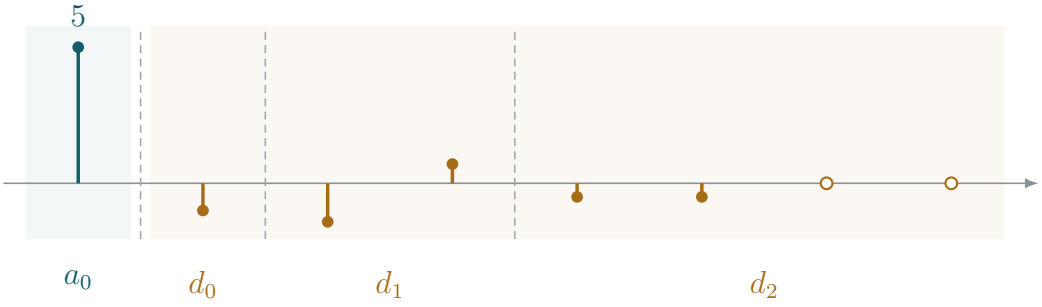


Figure 7.9: The eight coefficients from the table, grouped by resolution. Open circles denote zero detail coefficients, not discarded entries.

The first split, shown explicitly. Column m pairs fine indices $2m, 2m+1$ with the outputs $a_2[m], d_2[m]$.

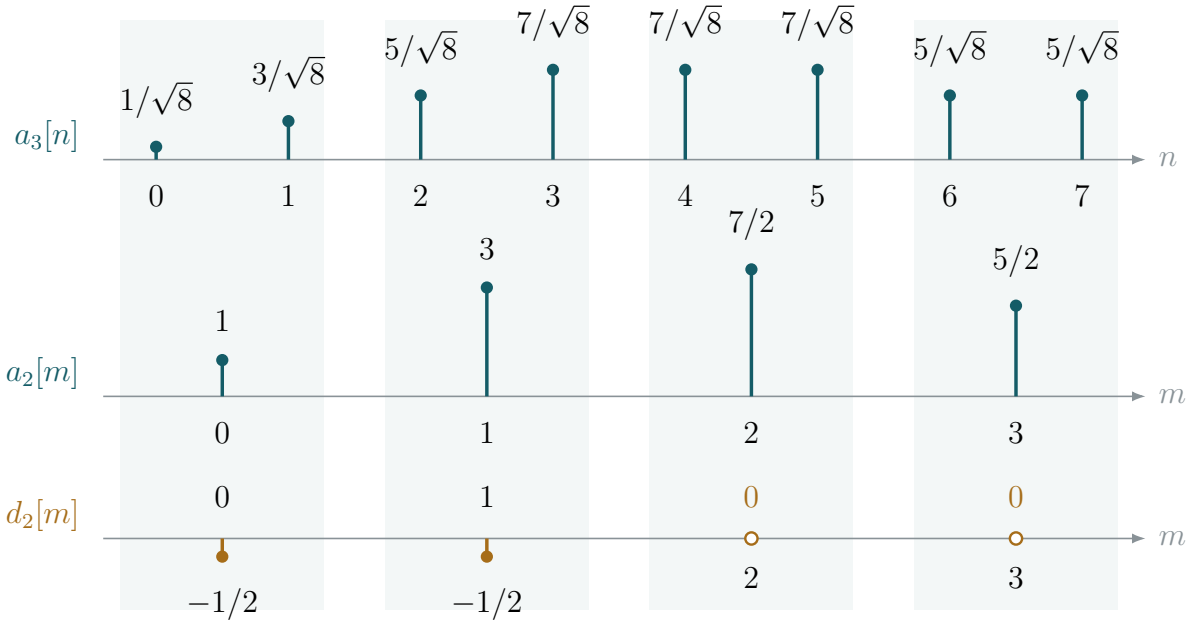


Figure 7.10: One input pair gives its sum and difference divided by $\sqrt{2}$. The rows share vertical units; open circles are zero detail coefficients.

Coefficients are not function heights. The scaling functions have height 2, so a_2 gives quarter-cell values $(2, 6, 7, 5)$. A detail coefficient $-1/2$ multiplies a wavelet of height $+2, -2$, restoring the values $-1, +1$:

$$q_2 = -\tfrac{1}{2}\psi_{2,0} - \tfrac{1}{2}\psi_{2,1}, \qquad f_3 = f_2 + q_2.$$

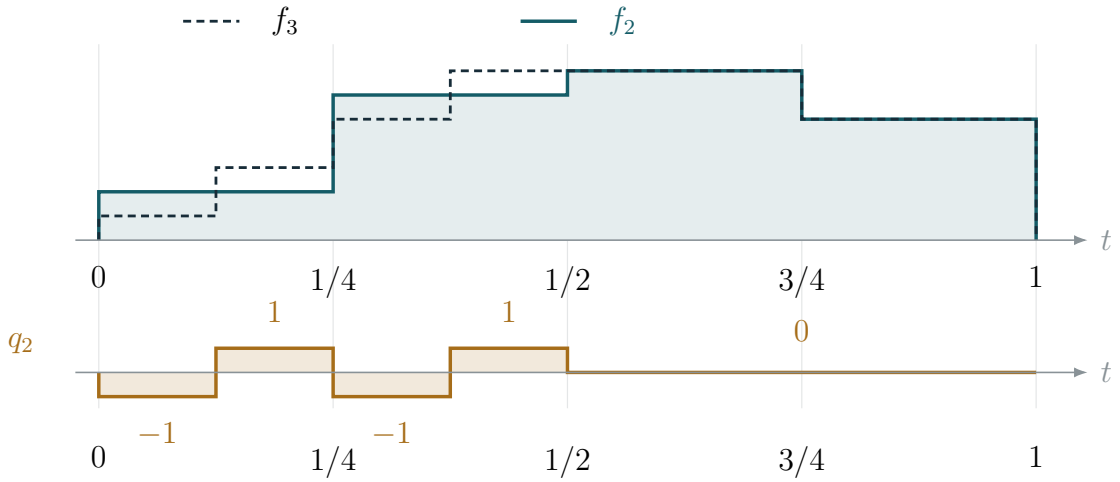


Figure 7.11: The same step in the signal domain. Above: f_3 (dashed) and f_2 (solid), whose quarter-cell heights are $2a_2 = (2, 6, 7, 5)$. Below: $q_2 = f_3 - f_2$, with eighth-cell values $(-1, 1, -1, 1, 0, 0, 0, 0)$. Both plots use the same vertical units.

7.14 Iterating only the approximation branch

Once a_3 has been split into a_2 and d_2, store d_2 and apply the same analysis bank to a_2. This produces a_1, d_1. One further split produces a_0, d_0.

The detail d_2 is not split again in this construction. It already describes the information added between V_2 and V_3. The next task is to separate the coarser information still grouped together in a_2.

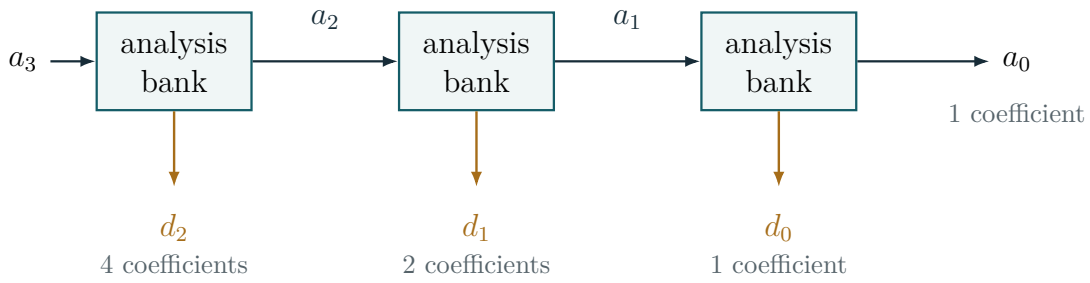


Figure 7.12: Three stages for an eight-coefficient input. The retained outputs have lengths $4, 2, 1, 1$, so their total remains eight. The arrows show computations on coefficient sequences, not new measurements of f.

More generally, start with a finest available sequence a_J and stop at a coarser level $J_0 < J$. The output is

$$a_{J_0}, \quad d_{J_0}, d_{J_0+1}, \ldots, d_{J-1}.$$

These sequences give the expansion

$$\boxed{P_J f = \sum_n a_{J_0}[n]\phi_{J_0,n} + \sum_{i=J_0}^{J-1} \sum_n d_i[n]\psi_{i,n}.} \tag{7.25}$$

Only levels $i < J$ are determined by a_J. To obtain d_J, we would need the additional information at resolution $J + 1$.

This cascade is the analysis half of a perfect-reconstruction filter bank: the recursive analysis procedure of the discrete wavelet transform (DWT).

The connection

Two-scale expansions become operations on coefficients. Convolution requires reversed analysis filters; the coarse grid requires downsampling. Keeping both branches preserves the acquired approximation, and repeating only the coarse branch organizes that information by resolution.